HERIOT WATT UNIVERSITY

School of Mathematical and Computer Sciences

Actuarial Mathematics and Statistics

F78AB

Actuarial and Financial Mathematics

IN-PERSON EXAMINATION

August 2024-2025

Duration: 2 Hours

Total marks available: 60

Answer all SEVEN questions

Resources Included within Exam Paper: No additional attachments
Additional Resources to be Supplied to Candidates: NIL
Calculator Permitted: YES

Ethical Commitment: By submitting answers to this paper, I declare that these answers are entirely my own work and have not been shared, in part or in whole or in any form, with any other student, or disseminated in any way.

I also confirm that I have reviewed and understand all University regulations, policies and procedures on assessment misconduct, including but not limited to accessing unauthorised materials during this examination or the use of unauthorised electronic devices.

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1. (a) Calculate the equivalent effective monthly discount rate if the annual effective interest rate is 5%.

[3 marks]

(b) Assume that there is an effective annual interest rate per annum, i, which remains unchanged over time. You are given that the present value of an n-year immediate annuity is 13, and that $\ddot{a}_{\overline{2n+1}|} = 23$.

(i) Determine the present value of a $2n$-year immediate annuity. [1 mark]

(ii) Use your answer from part (i) above to help you determine the value of v^n.

Hint : Consider the $2n$-year annuity as a combination of a regular annuity and a deferred annuity.

[3 marks]

[Total 7 Marks]

2. Julia takes out a £315,000 mortgage for 25 years at a nominal rate of 6.5% per year convertible monthly. Julia will be making regular monthly payments in arrears, commencing at the end of the month of receiving the loan.

(a) Show that Julia's level monthly payment is £2,126.90.

[2 marks]

(b) To pay off the loan faster, Julia decides to add £125 to each of the scheduled monthly payments, starting with the first repayment itself.

(i) Calculate the number of repayments Julia will make with this new payment arrangement.

[4 marks]

(ii) Calculate the amount of the final payment that Julia will make (ignoring the rounding error adjustment that is expected when a loan is amortised).

[4 marks]

[Total 10 Marks]

3. (a) A two-year investment requires £2m to be paid up-front and will then earn £1m and £3m at the end of years one and two respectively. With an effective annual interest rate of i, the investment has an NPV of 0. Calculate i.

[3 marks]

(b) Suggest three reasons why a company may decide not to invest in a project that has a positive NPV.

[3 marks]

(c) An investment fund was launched on 1 January 2022 with an initial investment of £500,000. Every six months after (ie on 1 July and 1 January each year), an additional £20,000 was added into the fund by investors.

The following fund balances were recorded:

30 June 2022	£540,000
31 December 2022	£550,000
30 June 2023	£580,000
31 December 2023	£620,000
30 June 2024	£650,000
31 December 2024	£710,000

(i) Show that the money-weighted rate of return earned by this fund during the period 1 January 2022 to 31 December 2024 is approximately 6.285% per annum.

[4 marks]

(ii) Calculate the time-weighted rate of return earned by this fund over the same period, 1 January 2022 to 31 December 2024.

[3 marks]

(iii) State, with brief reason, which of your answers to (i) or (ii) above is likely to be the better measure for comparing the performance of this fund with the performance of other investment funds.

[1 mark]

[Total 14 marks]

4. Maggie plans to purchase a deferred annuity from an insurance company. Maggie will receive 30 annual payments, beginning with £20,000 in five years time. Each annual payment will be 3.5% higher than the year before. Using an effective annual interest rate of 6%, calculate the purchase price of the annuity.

[Total 4 Marks]

5. The force of interest $\delta(t)$ is a function of time and at any time t (measured in years) is given by:

$$\delta(t) = 0.06 + 0.002t$$

(a) Calculate the value at time $t = 6$ of £1 invested at time 0. [2 marks]

(b) Calculate the value at time $t = 7$ of £1 invested at time 0. [2 marks]

(c) Calculate the value at time $t = 7$, of £1 invested at time 6. [1 mark]

[Total 5 Marks]

6. Neville purchases a bond six months after issue. The bond will be redeemed at par ten years after issue, and pays coupons of 7% per annum annually in arrears.

Neville pays tax of 20% on both income and capital gains.

(a) Calculate the purchase price of the bond per £100 nominal to provide Neville with a rate of return of 8% per annum effective. [6 marks]

(b) If Neville expects a real-rate of return of 2.5% per annum effective from the bond, calculate the annual rate of inflation he expects. [2 marks]

[Total 8 Marks]

7. An insurance company has a portfolio of annuity contracts under which it expects to pay £2 million at the end of each of the next 20 years, followed by £1 million at the end of the following 20 years.

The government bond with the longest duration in which the insurance company can invest its funds pays a coupon on 8% per annum in arrears, and is redeemed at par in 15 years time. The yield to maturity of the government bonds is 4% per annum effective and a coupon payment has just been made.

(a) Show that the duration of an n-year annuity of £1 *p.a.* paid annually in arrears is equal to

$$\frac{(Ia)^{i}_{\overline{n}|}}{a^{i}_{\overline{n}|}}$$

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[3 marks]

(b) Calculate the duration of the company's liabilities at a rate of interest of 4% per annum effective.

[4 marks]

(c) Calculate the duration of the insurance company's assets at a rate of interest of 4% per annum effective, if all the insurance company's funds are invested in the government bonds with the longest duration.

[3 marks]

(d) Explain why the insurance company cannot immunize its liabilities by purchasing government bonds.

[2 marks]

[Total 12 Marks]

END OF PAPER